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MATHEMATICS

TARGET : JEE (Main) 2024

CHAMP-2024 REVISION DPP NO.- 10

Syllabus:- Continuity & Derivability

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. If $f(x)=\begin{cases} ax+b & x\le 2 \\ x^2-5x+6, & 2<x<3 \\ px^2+qx+1, & x\ge 3\end{cases}$ is differentiable everywhere, then

(1) $a=-1,\ \ p=\frac{-4}{9}$ (2) $b=-2,\ \ q=-\frac{5}{3}$ (3) a = 1, b = 2 (4) $a=-1,\ \ q=\frac{5}{3}$

2. f(x) = $\begin{cases} x\dfrac{e^{\frac{1}{x}}-e^{\frac{-1}{x}}}{e^{\frac{1}{x}}+e^{\frac{-1}{x}}} & ,\ x\ne 0 \\ 0 & ,\ x=0\end{cases}$ then which of the following is true for f(x) at x = 0

(1) continuous and differentiable (2) continuous but not differentiable

(3) discontinuous (4) undefined

3. If f(x + y) = f(x) f(y) $\forall$x, y$\in$R and f(0) $\neq$ 0 then $\dfrac{f(x)}{1+(f(x))^2}$ is

(1) even (2) odd (3) neither even nor odd (4) > 1

4. Let f(x) = $||e^x-1|-1|$ then f(x) is differentiable at all points except at x =

(1) 0, 2 (2) ℓn2 (3) 1, ℓn2 (4) 0, ℓn2

5. If f(x) = $\begin{cases} \dfrac{1}{x^2}-\dfrac{1}{x^2}, & x<0 \\ \sin^{-1}(x+b), & x\ge 0\end{cases}$ Then which of the following statment about f(x) at x = 0 is false?

(1) Continuous if b = 0 (2) discontinuous for some b$\in$R

(3) differentiable for b = ± 1 (4) non-differentiable

6. Let f(x) = x|x|, g(x) = sin x and h(x) = (gof) (x). Then

(1) h(x) is not differentiable at x = 0

(2) h(x) is differentiable at x = 0, but h'(x) is not continuous at x = 0

(3) h'(x) is continuous at x = 0, but it is not differentiable x = 0

(4) h'(x) is differentiable at x = 0

7. Let f : $(-\infty,\infty)\to(-\infty,\infty)$ be given by f(x) = $x|x| + |x-1| - |x-2|^2 + |x-3|^3$. If A denotes the set of number of points where f(x) is not differentiable, then A =

(1) {3} (2) {1} (3) {0} (4) {2}

8. If f(x) = |x| + [x – 1], where [·] is greatest integer function, then f(x) is :

(1) continuous at x = 0 as well as at x = 1.

(2) Continous at x = 0 but not at x = 1

(3) continuous at x = 1 but not at x = 0

(4) neither continuous at x = 0 nor at x = 1

9. Amongst the following functions, a function that is differentiable at x = 0 is

(1) cos (|x|) – |x| (2) cos (|x|) + |x| (3) sin (|x|) + |x| (4) sin (|x|) – |x|

10. If f(x) is a differentiable function in the interval (0, ∞) such that f(1) = 1 and $\lim_{t\to x}\frac{t^2 f(x) - x^2 f(t)}{t-x} = 1$, for each x > 0, then $f\left(\frac{3}{2}\right)$ is equal to

(1) $\frac{25}{9}$ (2) $\frac{13}{6}$ (3) $\frac{23}{18}$ (4) $\frac{31}{18}$

11. If f(x) = $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$, then f(x) is differentiable on :

(1) [–1, 1] (2) R – {–1, 1} (3) R – (–1, 1) (4) None of these

12. If f(x) = $\begin{cases} (1+|\sin x|)^{a/|\sin x|}, & -\frac{\pi}{6} < x < 0 \\ b, & x = 0 \\ e^{\tan 2x/\tan 3x}, & 0 < x < \frac{\pi}{6} \end{cases}$ is continuous at x = 0, then -

(1) a = $\frac{1}{3}$, b = $e^{\frac{1}{3}}$ (2) a = $\frac{2}{3}$, b = $e^{\frac{2}{3}}$ (3) a = $\frac{1}{3}$, b = $e^{\frac{2}{3}}$ (4) a = $\frac{2}{3}$, b = $e^{\frac{1}{3}}$

13. If function f(x) = $\left(1+\frac{x}{p}\right)^{1/x}$; is continuous at x = 0 then f(0) is equal to

(1) e^p (2) e^{-p} (3) 0 (4) $e^{1/p}$

14. Function f(x) = $\begin{cases} x\sin x; & 0 \le x < \pi/2 \\ \frac{\pi}{2}\sin(\pi + x); & \frac{\pi}{2} \le x < \pi \end{cases}$ then -

(1) $\lim\limits_{x\to\pi/2}$ does not exist (2) $\lim\limits_{x\to\pi/4}$ does not exist

(3) Discontinuous at x = π/2 (4) Both (A) & (C)

15. The point of discontinuity in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ of f(x) = $\frac{1+\cos 5x}{1-\cos 4x}$ is

(1) 0 (2) π/4 (3) –π/4 (4) None of these

16. Let f and g be differentiable functions satisfying g′(a) = 2, g(a) = b and $f$og = I (identity function). Then, f'(b) is equal to –

(1) 2 (2) 2/3 (3) 1/2 (4) None of these

17. If f(x) is continuous and increasing function such that the domain of g(x) = $\sqrt{f(x)-x}$ be R and h (x) = $\frac{1}{1-x}$ then the domain of ϕ(x) = $\sqrt{f(f(f(x))) - h(h(h(x)))}$ is -

(1) R (2) {0, 1} (3) R – {0, 1} (4) R^+ – {1}

18. Function $\begin{cases} 2x\tan x - \dfrac{\pi}{\cos x}; & x \neq \dfrac{\pi}{2} \\ k; & x = \dfrac{\pi}{2} \end{cases}$ is continuous at x = $\dfrac{\pi}{2}$ if k =

(1) – 2 (2) 2

(3) $\frac{1}{2}$ (4) no such values of k exists

19. The value of f(0) so that f(x) = $\dfrac{-e^x + 2^x}{x}$ is continuous at x = 0, will be

(1) – ℓn 2 (2) 0 (3) 4 (4) $\ell n\left(\frac{2}{e}\right)$

20 y = $\dfrac{1}{t^2+t-2}$ where t = $\dfrac{1}{x-1}$, then the number of points of discontinuities of y = f(x), x $\in$ R is-

(1) 1 (2) 2 (3) 3 (4) Infinite

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. The number of points of non-differentiability of f(x) = ||x| –1| + | cos πx| for – 2 < x < 2 is

22. If function f(x) = $\dfrac{\sin x}{\sin\frac{1}{x}}$ is discontinuous at k points in $\left(\dfrac{1}{2012\pi}, \dfrac{1}{2000\pi}\right)$ then the value of k is

23. Let f(x) and g(x) be differentiable functions on (–∞,∞) and let f '(x) and g '(x) denote derivatives of f(x) and g(x) respectively. If f (0) = $\frac{1}{2}$, g(0) = $\frac{1}{3}$, f ' (0) = 1 and g ' (0) = 2, then the value of

$\lim\limits_{x\to 0} \dfrac{2f(2x^2 = 3x) - 1}{3g(x) - 1}$ is

24. If f(x) is a twice differentiable function such that f ''(x) = –f(x), f '(x) = g(x), h(x) = $f^2(x) + g^2(x)$, h(5) = 11 then h'(10) =

25. The number of points of discontinuity of f(x) = [x^3 +1] in (1, 2) is/are

ANSWER KEY_DPP-10

1.	**(2)**	**2.**	**(2)**	**3.**	**(1)**	**4.**	**(4)**	**5.**	**(3)**	**6.**	**(3)**	**7.**	**(2)**
8.	**(4)**	**9.**	**(4)**	**10.**	**(4)**	**11.**	**(2)**	**12.**	**(2)**	**13.**	**(4)**	**14.**	**(4)**
15.	**(1)**	**16.**	**(3)**	**17.**	**(3)**	**18.**	**(1)**	**19.**	**(4)**	**20.**	**(3)**	**21.**	**(7)**
22.	**(11)**	**23.**	**(1)**	**24.**	**(0)**	**25.**	**(6)**						